Quang Truong

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Research Interests

Machine Learning, Graph Neural Networks, Geometric Deep Learning, Relational Deep Learning, Generative AIs

Education

Michigan State University — *East Lansing, MI*
Ph.D. in Computer Science — *Aug. 2024 - Present*

Dartmouth College — *Hanover, NH*
M.S. in Engineering Sciences — *Sep. 2022 - Jun. 2024*

Selected Publications

‡ indicates equal contribution.

[1] TopoX: A Suite of Python Packages for Machine Learning on Topological Domains
Mustafa Hajij‡, Mathilde Papillon‡, Florian Frantzen‡, ... **Quang Truong**, and ... Nina Miolane. **2024a**
Under review at the Journal of Machine Learning Research (JMLR).

[2] Weisfeiler and Lehman Go Paths: Learning Topological Features via Path Complexes
Quang Truong and Peter Chin. **2024b**
AAAI'24: The 38th Annual AAAI Conference on Artificial Intelligence.

[3] Not All Data Matters: An Efficient Approach to Multi-Domain Learning in Vehicle Re-Identification
Quang Truong and Bo Mei. **2021**
ITSC'21: The 24th IEEE International Conference on Intelligent Transportation Systems.

Skills

Programming	Python, C/C++, Java, SQL, R, MatLab
Machine Learning	Pytorch, Pytorch Geometric, Pytorch Frame, NetworkX, TopoX, Pytorch Lightning, Scikit-learn, Tensorflow, Keras, Matplotlib, Numpy, Pandas
DevOps	WandB, Hydra, Docker, Singularity, Pytest, AWS, Google Cloud

Research Experience

DSE Lab, Michigan State University — *East Lansing, MI*
*Ph.D. Student (Advisor: **Jiliang Tang**)* — *Aug. 2024 - Present*
- Utilizing Graph Neural Networks (GNNs) to model and learn relationships between entities in relational databases.
- Bridging the gaps between GNNs and propositionalization-based approaches for deep tabular learning.

pyt-team — *Remote*
Contributor — *Aug. 2023 - Oct. 2023*
- Spearheaded the development of Path Complex package for TopoNetX library, which provides reliable and user-friendly building blocks for computing and machine learning on topological domains that extend graphs.

LISP Lab, Dartmouth College — *Hanover, NH*
*M.S. Student (Advisor: **Peter Chin**)* — *Sep. 2022 - Jun. 2024*
- Introduced a novel approach to Topological Deep Learning (TDL) leveraging simple paths, significantly enhancing graph expressivity and surpassing the theoretical limitations of the 3-WL test.
- Developed Path Isomorphism Network, a state-of-the-art TDL model, which delivered outstanding performance on benchmark datasets: PROTEINS (78.8%), NCI1 (85.1%), NCI109 (84.0%), and IMDB-B (76.6%).

Vision Lab, University of Illinois at Chicago — *Remote*
*Research Intern (Advisor: **Wei Tang**)* — *May 2021 - Feb. 2022*
- Designed a context-aware 3D object detection architecture that captures both object-wise and object-context relationships, improving spatial understanding.
- Boosted model performance through cross-modality alignment, improving the feature representation of objects.

ML Lab, Texas Christian University *Fort Worth, TX*
*Undergraduate Research Assistant (Advisor: **Bo Mei**)* *May 2019 - May 2021*

- Developed an advanced Vehicle Re-identification pipeline utilizing Generative Adversarial Networks (GANs) for adaptive domain learning, resulting in a 10% improvement in mean Average Precision (mAP) over the baseline.
- Engineered a novel filtering algorithm that reduces the training data requirement by 25% while maintaining model performance, significantly improving computational efficiency.

Presentations

Invited Presentations

- Topologically-driven Neural Networks, *HPCC Special Seminar: Decoding Computational Challenges, the Advanced Institute of Interdisciplinary Science and Technology - HCMUT, VNU-HCM*. **2024**
- Domain-invariant Network for Vehicle Re-identification, *Annual Industrial Board Meeting of TCU Department of Computer Science*. **2020**

Conference Presentations

- Weisfeiler and Lehman Go Paths: Learning Topological Features via Path Complexes, *AAAI'24: The 38th Annual AAAI Conference on Artificial Intelligence*. **2024**
- Not All Data Matters: An Efficient Approach to Multi-Domain Learning in Vehicle Re-identification, *ITSC'21: The 24th IEEE International Conference on Intelligent Transportation Systems*. **2021**

Teaching Experience

Dartmouth College

- Teaching Assistant, *ENGS 96: Mathematics for Machine Learning*. **Winter 2024**

Honors

Grants

- Research Grant for Domain-invariant Network for Vehicle Re-identification ($1471), *TCU SERC Undergraduate Research Grant*. **2020**
- Research Grant for AI-2-Go ($1500), *TCU SERC Undergraduate Research Grant*. **2019**
- Research Grant for Housing Price Prediction ($1500), *TCU SERC Undergraduate Research Grant*. **2019**

Awards

- VEF Fellow, *VEF 2.0 Program, VEF Fellows and Scholars Association*. **2021**
- Vingroup Scholarship Recipient, *Master's & Ph.D. Scholarship Program, VinUniversity*. **2021**
- Best Undergraduate Research Poster, *TCU Student Research Symposium [URL]*. **2019, 2021**
- TCU Scholar, *Texas Christian University*. **2020, 2021**
- Dean's List, *Texas Christian University*. **2019-2021**
- Bronze Medal, *ACM-ICPC South Central USA Regional Contest [URL]*. **2019**
- 1st Prize, *Calculus Bee [URL]*. **2019**
- Transfer Faculty Scholarship ($196,000 for four years), *Texas Christian University*. **2019**
- President's Scholar, *Mississippi State University*. **2017, 2018**
- Freshmen Academic Excellence Scholarship ($54,000 for four years), *Mississippi State University*. **2017**

Relevant Coursework

Engineering	Signal Processing, Fourier Transforms and Complex Variables, Statistical Methods in Engineering
Computer Science	Artificial Intelligence, Multi-modalities Generative AI, Network Science and Complex Systems, Deep Learning, Data Mining and Visualization, Analysis of Algorithm, Operating Systems, Database Systems
Mathematics	Mathematics for Machine Learning, Game Theory, Applied Linear Algebra, Probability, Statistics, Discrete Mathematics, Calculus